Crossing Is Not Insertion: Recoverability-Aware Evaluation in 2D Orbital Control

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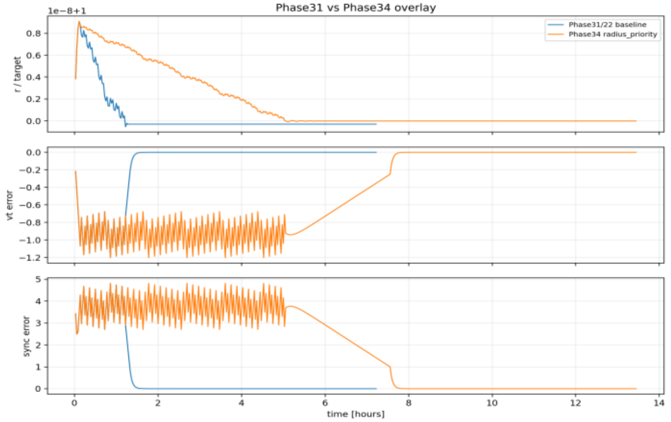
Q1: Research Question

Is target-radius crossing sufficient to indicate recoverability?

A crossing event can occur before the post-cross state is dynamically recoverable.



Recoverable crossing = crossing + later basin entry

Crossing = reaching target radius
Recoverability = later entering a simulator-defined basin

Scope: simplified 2D simulator; no real-flight validation.

Q2: Methodology / Project Design

Typed workflow keeps the method readable in one glance.

Data — 2D simulator, 24 cases → **Rollouts** — controllers + PPO / IL → **Metrics** — crossing, recoverable → **Evidence** — compare + validate

DATA
Simulator-generated state-action rollouts

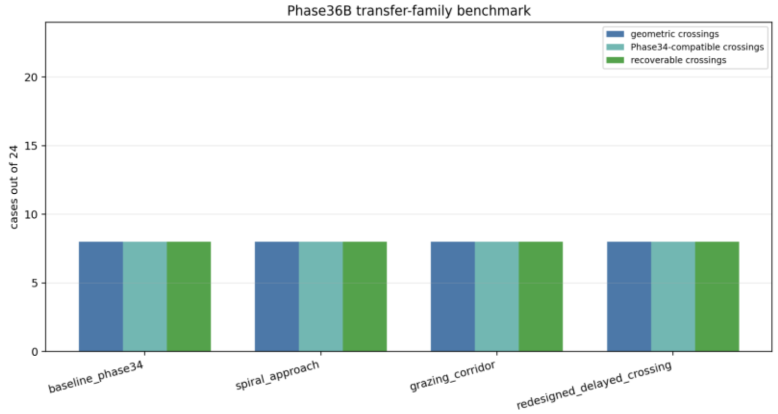
BENCHMARK
3 radii × 4 angles × 2 thrust scales

AI BASELINES
PPO and imitation learning tested

All evidence is reported with simulator-defined scope and stored artifacts.

Q3: Data Analysis & Results

Strongest aggregate result: upstream variants did not expand the crossing basin.



Key evidence chain

Phase31 → 34
Crossing: 8/24 → 8/24

Phase31 → 34
Recoverable: 0/24 → 8/24

Phase36B
Each family: 8/24 crossings + 8/24 recoverable crossings, overspeed = 0, Instability = 0

Phase37 diagnostics
144 rollouts: 0/16 new subset: 0/4 selected

Q4: Interpretation & Conclusions

ANSWER
Target-radius crossing is not sufficient for simulator-defined recoverability.

MAIN FINDING
Post-cross synchronization made existing crossing-producing cases recoverable.

LIMITATION
Simplified 2D dynamics; fixed thresholds; limited 24-case grid; no flight validation.

NEXT STEP
Improve upstream crossing generation with evidence-ranked controller search.

PPO and imitation-learning baselines were investigated, but did not provide the main positive result.